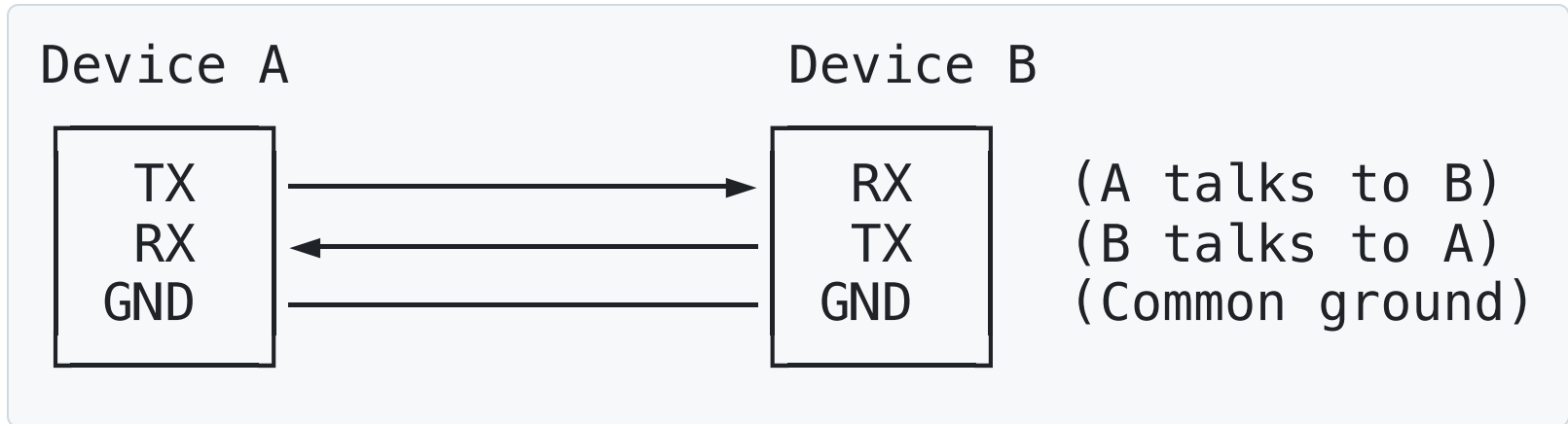


RS-232: The Phone Call Protocol

Basic Concept:



Like a phone conversation:

- **TX (Transmit)** = Your mouth (speaking)
- **RX (Receive)** = Your ear (listening)
- **GND** = Same language/understanding

UART = Universal Asynchronous Receiver/Transmitter

RS-232 = Recommended Standard 232 (voltage level specification)

- No slave/master, just two equal devices talking
- Asynchronous (no clock signal, like talking without a metronome)
- Long-distance capable (up to 15m for RS-232)
- Simple and universal (used in GPS, Bluetooth, WiFi modules, sensors)

RS-232 Protocols

- UART
- Voltage Levels
- MAX232 Level Converter and USB-to-Serial
- Data Format
- Configuration

UART (Universal Asynchronous Receiver/Transmitter)



Important: TX of Device A → RX of Device B
RX of Device A ← TX of Device B
(Crossover connection!)

Key points:

- **TX** (Transmit): Sends data OUT
- **RX** (Receive): Receives data IN
- **GND** (Ground): Common reference (essential!)
- **Cross-connect:** TX → RX, RX → TX
- **No clock signal:** Asynchronous, timing based on agreed baud rate

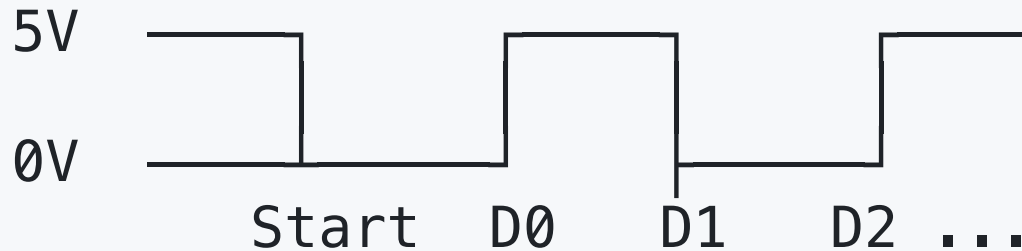
Voltage Levels

TTL Serial (Modern microcontrollers)

Logic HIGH (1): +3.3V or +5V

Logic LOW (0): 0V (GND)

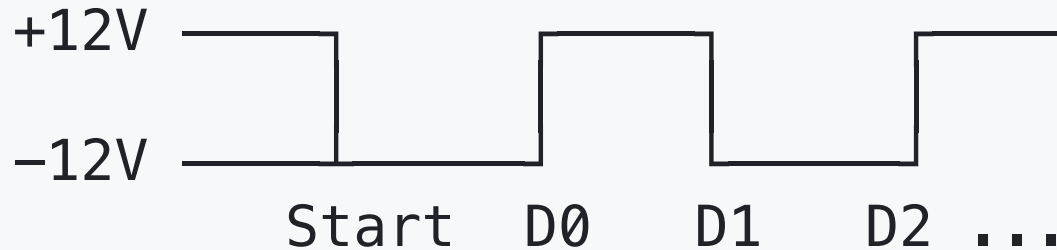
Voltage:



RS-232 (Traditional computer serial ports)

Logic HIGH (1): -3V to -15V (typically -12V)
Logic LOW (0): +3V to +15V (typically +12V)

Voltage:

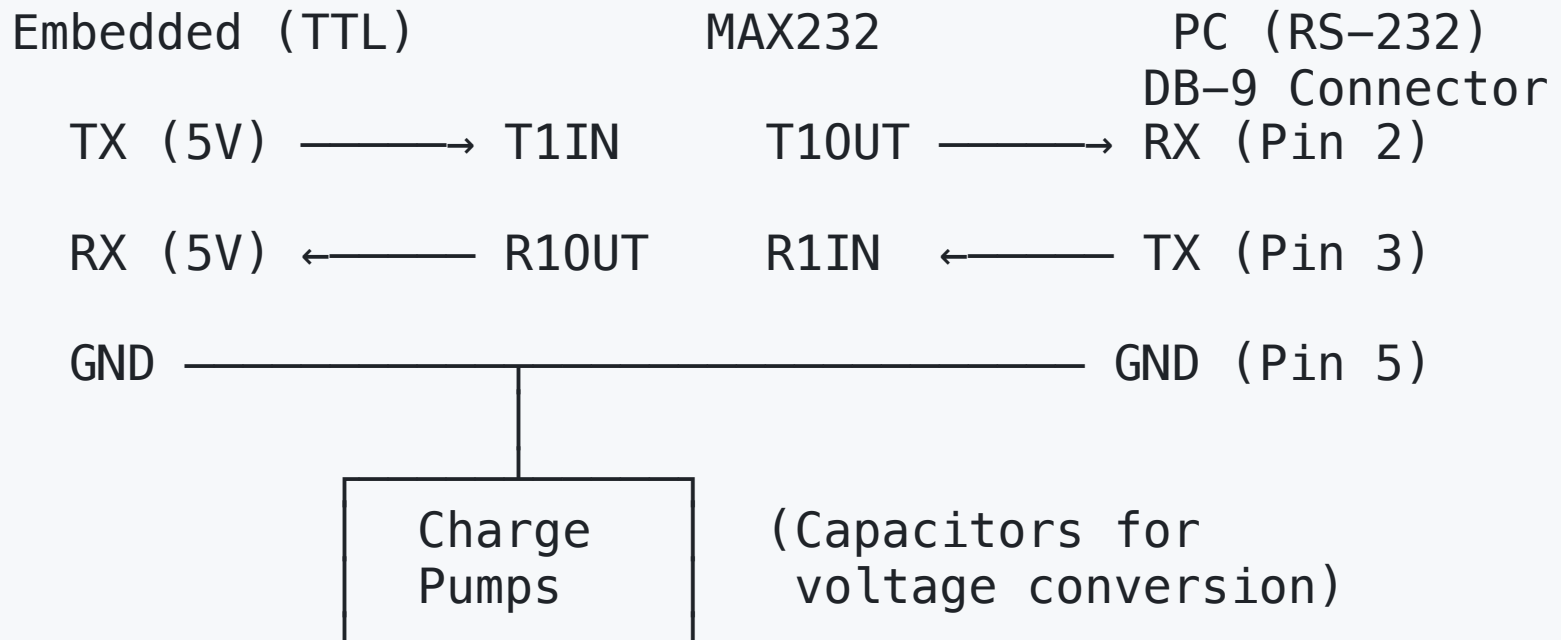


Note: Inverted logic compared to TTL!

Important: Need a level converter (MAX232, MAX3232) to connect TTL to RS-232!

MAX232 Level Converter and USB-to-Serial

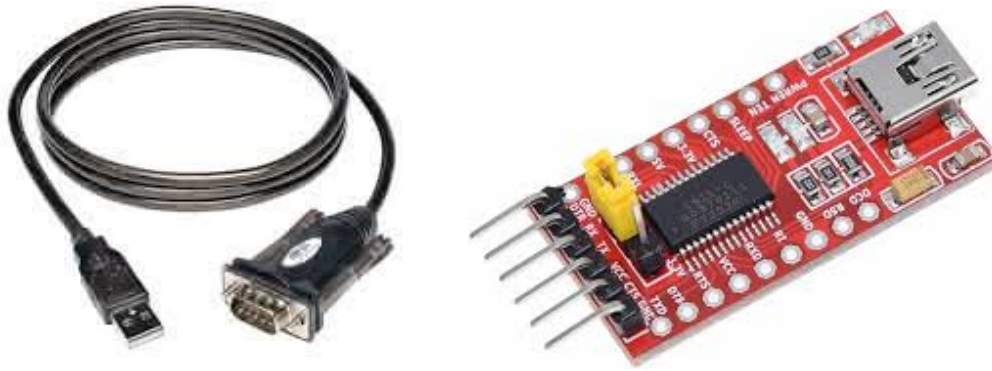
Connecting Embedded System to PC serial port (RS-232):



MAX232 chip:

- Converts 5V TTL $\rightarrow$ $\pm 12\text{V}$ RS-232
- Converts $\pm 12\text{V}$ RS-232 $\rightarrow$ 5V TTL
- Requires 4-5 external capacitors ($1\mu\text{F}$)
- Allows long-distance communication

Most modern PCs don't have RS-232 ports. Instead, we use USB-to-Serial adapters that emulate RS-232 over USB.



Arduino uses USB-to-Serial internally (FTDI, CH340, or CP2102 chip or similar) to program and communicate with the board.

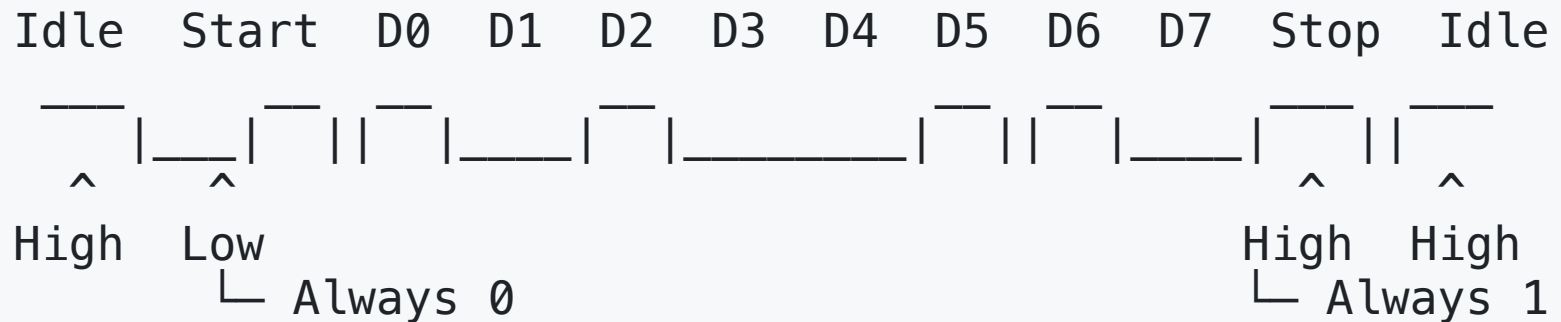
Why Weird Voltages?

- Noise immunity - larger voltage swings
- Long cable runs - can travel further
- Historical reasons - from telegraph era

Modern Reality:

Most "RS-232" today actually uses **3.3V UART** (same concept, normal voltages)

Data Format



Sending 'A' (ASCII 65 = 01000001):

1. Idle = High voltage (no data)
2. Start bit = Low (get ready!)
3. 8 data bits = 01000001 (LSB first: 10000010)
4. Stop bit = High (finished!)
5. Idle = High (waiting for next character)

Configuration

Communication Settings (like dialing instructions):

```
# Both devices must agree on these settings!
baud_rate = 9600      # bits per second (like talking speed)
data_bits = 8         # usually 8 (sometimes 7)
parity = None         # error checking (None/Even/Odd)
stop_bits = 1         # end-of-character marker
flow_control = None   # traffic control (None/Hardware/Software)
```

Common Configurations:

- **9600,8,N,1** = 9600 baud, 8 data bits, no parity, 1 stop bit
- **115200,8,N,1** = High-speed version

Both devices must use EXACTLY the same settings!

Why UART/RS-232?

Advantages:

- **Simple:** Only 2 wires (TX and RX)
- **Well-established:** Decades of proven use
- **Long-distance:** Up to 15m (RS-232) or 1km+ (RS-485)
- **Universal support:** Every microcontroller has UART
- **Human-readable:** Can send ASCII text easily

Common uses:

GPS Module	→ Send location data
Bluetooth Module	→ Wireless communication
GSM/4G Module	→ Cellular communication
Computer Debug	→ Serial monitor/terminal
Arduino <-> PC	→ Programming and debugging
Sensor Networks	→ Simple data exchange

Timing and Synchronization

Calculation:

At 9600 baud with 8N1 (10 bits per byte):
 $9,600 \text{ bits/sec} \div 10 \text{ bits/byte} = 960 \text{ bytes/sec}$

Sending "Hello":
 $5 \text{ bytes} \times 10 \text{ bits/byte} = 50 \text{ bits}$
 $\text{Time} = 50 \text{ bits} \div 9,600 \text{ bps} = 5.2 \text{ milliseconds}$

Important: Both transmitter and receiver **MUST** use the same baud rate!

Transmitter @ 9600 baud:

Each bit = $1/9600 = 104$ microseconds

104	104	104	104	104	104	104	104	μs per bit
-----	-----	-----	-----	-----	-----	-----	-----	------------

Receiver @ 9600 baud: ✓ Correctly receives data

Receiver @ 19200 baud: x Samples twice per bit (wrong!)

Receiver @ 4800 baud: x Samples every 2 bits (wrong!)

Tolerance:

- Most UARTs tolerate $\pm 5\%$ baud rate mismatch
- Beyond that: garbled data
- Use accurate crystal oscillator (not internal RC)